CMPE 255 · SHORT STORY

Mining 500,000 Steam Reviews

What data mining reveals about player perceptions of generative AI in games

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Based on Bazzaz & Cooper (2026)

Research presentation | 10–15 minutes

RESEARCH QUESTION

Two forms of automation. Different reactions.

The study compares perceptions, not equivalent technologies.

Procedural generation (PCG)

Rules and algorithms create worlds, levels, encounters, and replayability.

Often experienced directly as a gameplay feature.

Generative AI

Learned models produce assets such as images, voice, text, and dialogue.

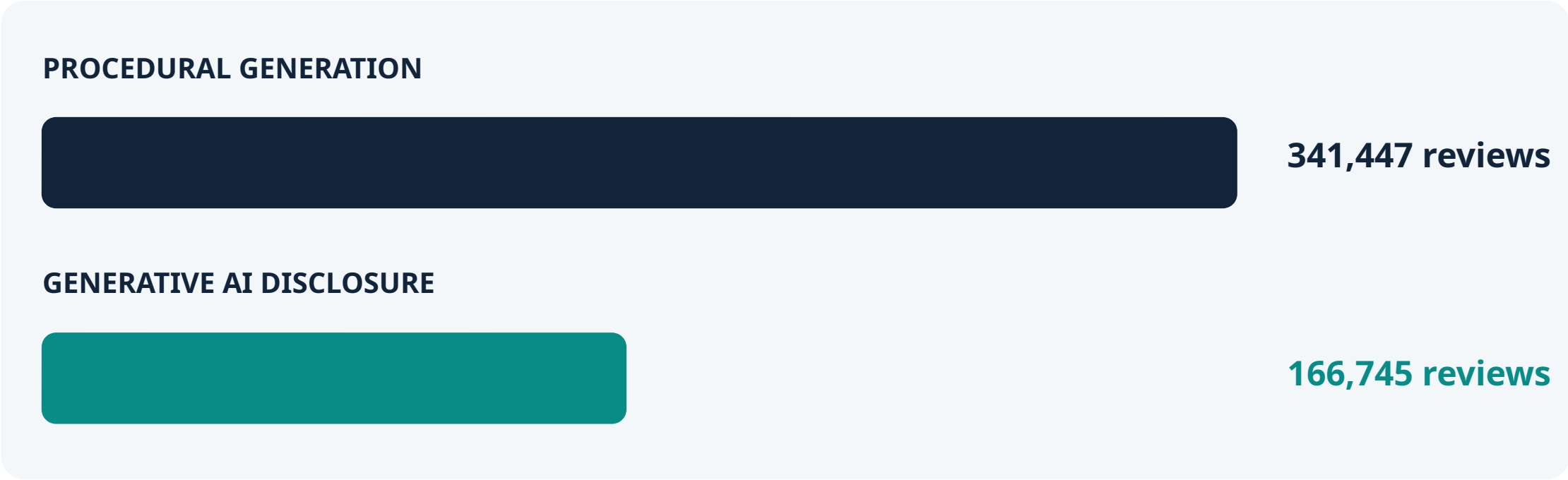
Often noticed through disclosures or visible content.

Research question: how do Steam reviews differ?

STUDY DESIGN

508,192 English-language Steam reviews

Games released between 2010 and 2025; two groups identified on Steam.

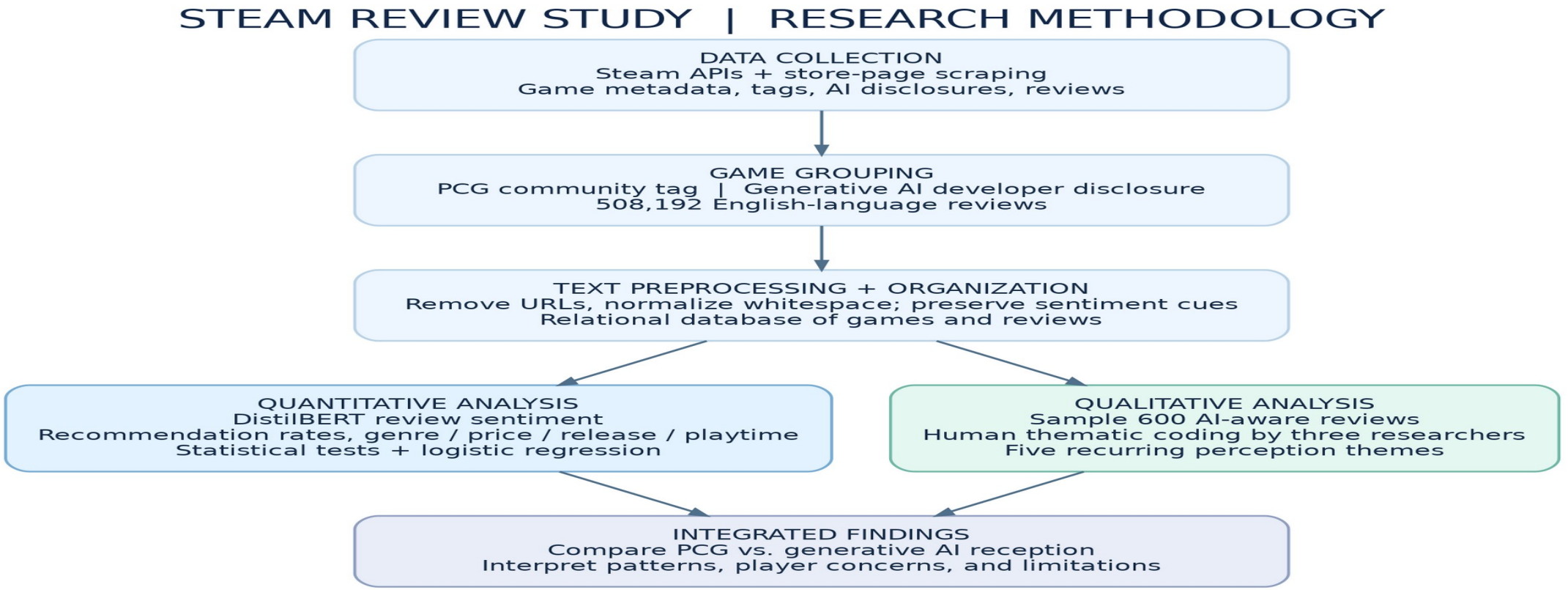


5,186 games tagged PCG

5,970 games disclosing AI use

The mixed-methods research pipeline

Quantitative mining + qualitative analysis of AI-aware reviews.



How DistilBERT reads review sentiment

A compact, transformer-based NLP model processes large volumes of text.

Figure 3 | DistilBERT sentiment workflow

Simplified application pipeline reconstructed from Bazzaz & Cooper (2026), Section 3.2



What the study actually reports

The model analyzes review language; researchers separately compare its sentiment classifications with Steam thumbs-up recommendations.

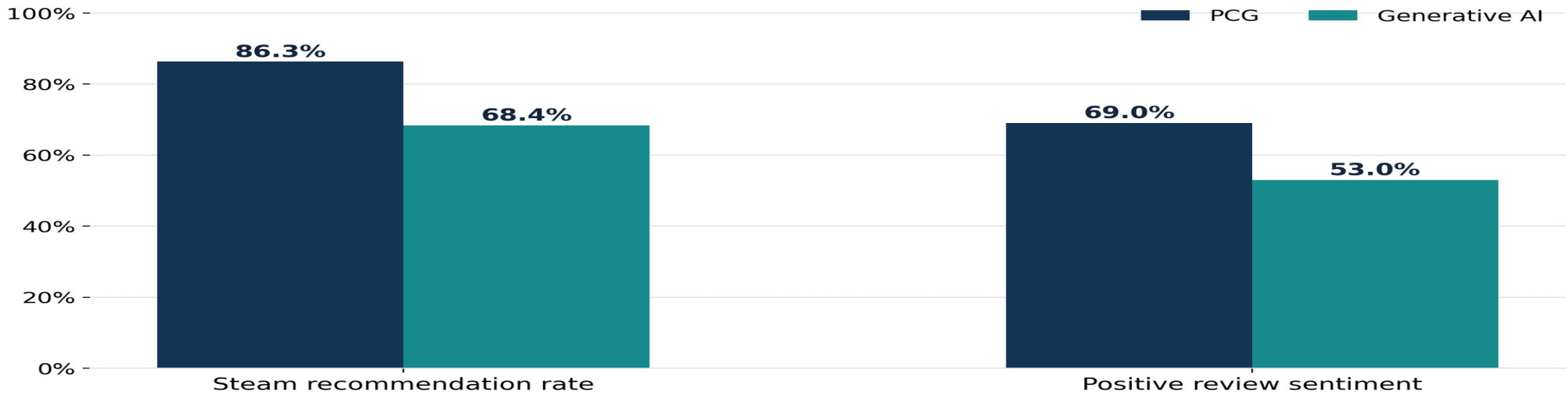
This is a conceptual workflow, not an original figure from the paper or a claim about the model’s internal training procedure.

A consistent gap in reception

Descriptive results: PCG versus games disclosing generative AI.

Figure 4 | Review outcomes

Steam reviews: procedural generation (PCG) vs. generative AI disclosures

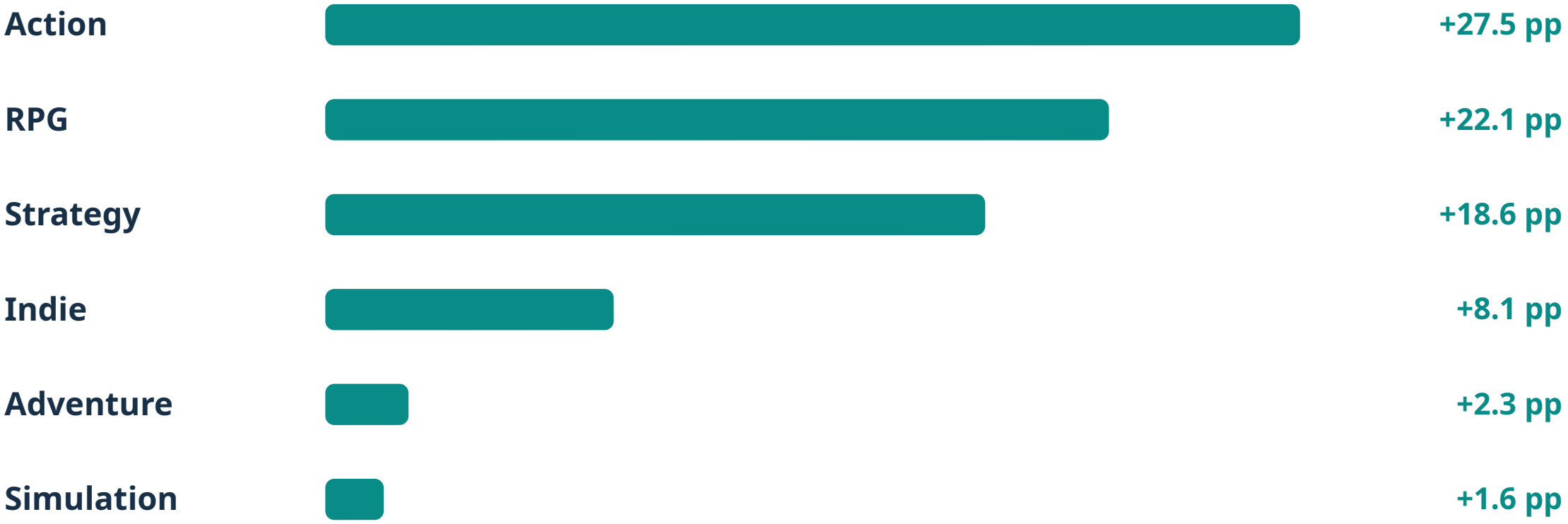


Source: Bazzaz & Cooper (2026), Table 2. Observational comparison; not a causal estimate.

SUBGROUP ANALYSIS

The gap depends strongly on game genre

Recommendation-rate difference: PCG minus generative AI, percentage points.



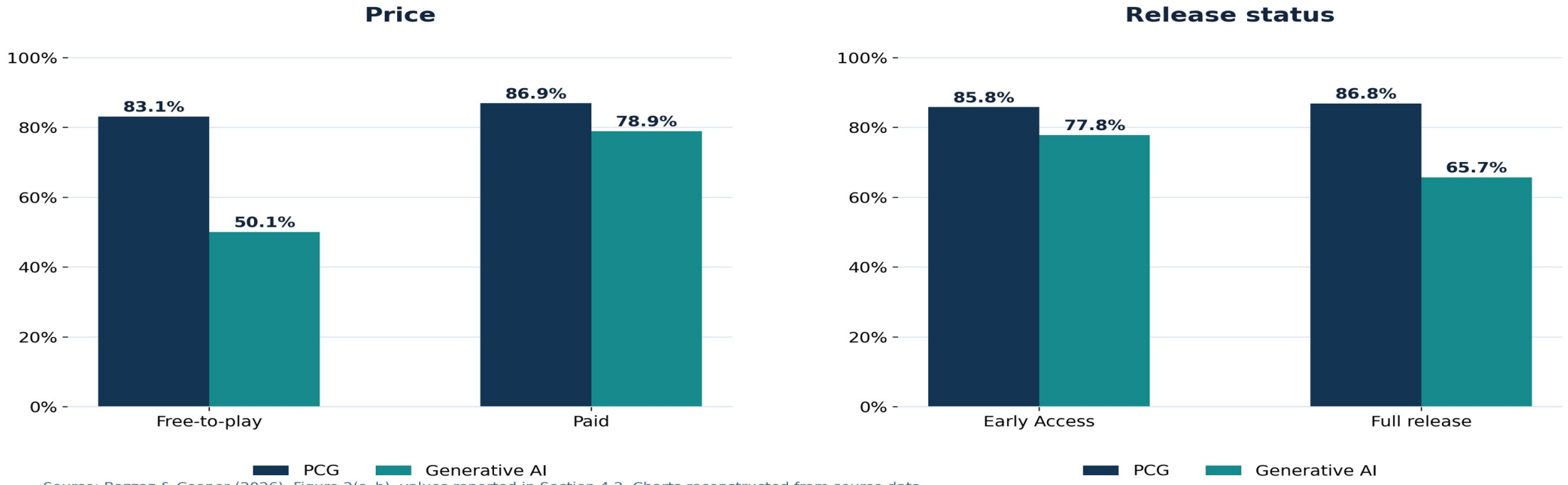
Selected genres from Table 4; a larger bar means a larger PCG advantage.

Price and release status change the pattern

Recommendation rates reconstructed from the paper’s Figure 2.

Figure 5 | Price and release status

Recommendation rates reported for PCG games and games disclosing generative AI



Source: Bazzaz & Cooper (2026), Figure 2(a–b), values reported in Section 4.2. Charts reconstructed from source data.

Five themes explain the player language

Three researchers thematically coded a sample of 600 AI-aware reviews.

Figure 6 | Five themes in player perceptions

Qualitative coding of 600 AI-aware Steam reviews; adapted from Bazzaz & Cooper (2026), Table 5

01	**Perceived quality deficit**	AI-made assets are interpreted as poor quality, low effort, or lack of developer care.
02	**Ideological and ethical resistance**	Objections involving artists, training data, and the broader effects of AI adoption.
03	**Acceptance of useful AI**	Support when AI enables enjoyable gameplay or clearly adds something new.
04	**Transparency and trust**	Negative reactions when AI disclosures appear incomplete or misleading.
05	**The bar keeps rising**	Expectation that AI should improve quality and experience, not merely lower costs.

Themes are qualitative categories, not sequential steps. No prevalence percentages are implied.

Controlling for measurable confounders

Logistic regression tests whether the reception gap remains after adjustments.

0.63

adjusted odds ratio

Generative AI group vs. PCG

95% CI: 0.62–0.64 • p < .001

Included controls

Free-to-play status
Early Access status
Playtime
Source × price interaction

What the data can—and cannot—say

Observational data mining is powerful, but not a randomized experiment.

What we can say

• The two game groups have different observed recommendation and sentiment patterns.

• The gap varies across genre, price, release status, and playtime.

What remains uncertain

• Whether AI usage itself causes lower ratings.

• Effects of overall game quality, marketing, developer reputation, and reviewer selection.

TAKEAWAYS

The broader lesson: technology must add player value

Three implications for developers and for data mining research.

01 · Value over novelty

AI is received more positively when it improves the experience instead of only cutting production effort.

02 · Trust matters

Clear disclosure and human review of outputs can shape perceptions of quality and care.

03 · Mixed methods win

Sentiment scores reveal patterns; qualitative themes help explain what those patterns may mean.

Discussion prompt: would your opinion change if you learned a favorite game used generative AI?

Research references

Key sources for the study, sentiment analysis, and related work.

PRIMARY STUDY

Bazzaz, M., & Cooper, S. (2026). Player Perceptions of Generative AI in Games: A Steam Review Analysis.

https://arxiv.org/abs/2608.11539

MODEL BACKGROUND

Sanh, V., et al. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter.

https://arxiv.org/abs/1910.01108

RELATED WORK

See the primary paper's Related Work section for Steam-review mining and AI disclosure studies.

https://arxiv.org/html/2608.11539v1

All figures are original reconstructions or adaptations of reported methods and results.